AUTOMATED RESUME ANALYZER FOR JOB PORTALS

AN INTERNSHIP REPORT

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Submitted by

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chapter-1: Abstract

The Automated Resume Analyzer for Job Portals is an intelligent recruitment system designed to simplify and automate the process of resume screening. In today's competitive job market, organizations receive a large number of resumes for every job opening, making manual evaluation both time-consuming and prone to human error. This project addresses these challenges by utilizing Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) techniques to analyze resumes automatically and compare them with job descriptions.

The system extracts essential information such as candidate name, contact details, educational qualifications, technical skills, certifications, and work experience from resumes submitted in PDF or DOCX formats. After preprocessing the extracted text, the system identifies relevant keywords and compares them with the required skills mentioned in the job description. Using similarity algorithms, the application calculates a matching score and ranks candidates according to their suitability for the job. This automated process significantly reduces recruitment time, minimizes bias, improves hiring accuracy, and enables recruiters to identify qualified candidates more efficiently.

1.2 : Introduction

Recruitment plays a vital role in determining the success of any organization because selecting the right employees directly influences productivity and business growth. With the increasing popularity of online job portals such as LinkedIn, Indeed, Naukri, and Monster, recruiters receive hundreds or even thousands of resumes for a single job vacancy. Reviewing such a large number of applications manually requires considerable time and effort, often resulting in delayed hiring decisions and the possibility of overlooking qualified candidates.

The Automated Resume Analyzer for Job Portals is developed to overcome these challenges by introducing automation into the recruitment process. The system uses Artificial Intelligence and Natural Language Processing to understand resume content and compare it with job requirements. Instead of manually reviewing every application, recruiters receive a ranked list of candidates based on their qualifications and skills. This approach improves the speed, accuracy, and fairness of the hiring process while reducing operational costs.

Chapter-2 : Problem Statement

Many organizations continue to rely on manual resume screening despite the rapid growth in online job applications. Recruiters often spend several hours reviewing resumes, comparing candidate qualifications with job requirements, and preparing shortlists for interviews. This process becomes increasingly difficult as the number of applications grows. Human fatigue, unconscious bias, and inconsistent evaluation criteria can result in suitable candidates being overlooked. Additionally, manual screening increases recruitment costs and delays the hiring process. Therefore, there is a need for an intelligent system capable of automatically analyzing resumes, matching them with job descriptions, and ranking candidates according to their suitability.

Chapter-3 : Objectives

The primary objective of this project is to develop an intelligent resume analysis system that automates candidate screening and improves recruitment efficiency. The system aims to extract relevant information from resumes, identify technical and professional skills, compare candidate profiles with job descriptions, and calculate a similarity score based on predefined criteria. Another important objective is to reduce manual effort, eliminate bias during candidate selection, improve hiring accuracy, and provide recruiters with a faster and more reliable method for identifying suitable applicants.

Chapter -4 : Literature Survey

Title: ***Survey on Resume Parsing Models for JOBCONNECT+: Enhancing Recruitment Efficiency using Natural Language Processing and Machine Learning***
Authors: R. Deepa, V. Jayalakshmi, K. Karpagalakshmi, S. Manikanda Prabhu, P. Thilakavathy
Year: 2025

This survey paper reviewed various resume parsing models used in modern recruitment systems. The authors discussed rule-based approaches, machine learning techniques, and deep learning models for extracting candidate information from resumes. The study highlighted that NLP-based resume parsing significantly reduces recruiter workload and improves hiring efficiency. However, it also identified challenges such as handling different resume formats and accurately extracting skills from unstructured documents.

Chapter-5 : Existing System

In the existing recruitment process, candidates submit their resumes through online job portals or company websites. Human resource personnel manually download each resume and review the candidate's educational background, technical skills, work experience, and certifications. The recruiter then compares this information with the job requirements before preparing a shortlist for interviews. Although this method has been widely used for many years, it suffers from several limitations. Manual screening is slow, labor-intensive, inconsistent, and susceptible to human error. As the number of applications increases, recruiters may unintentionally overlook highly qualified candidates or spend excessive time evaluating resumes that are not relevant to the job position.

Chapter-6 : Proposed System

The proposed Automated Resume Analyzer introduces an intelligent and automated approach to resume screening. Candidates upload their resumes through a web-based application, after which the system extracts textual information from the uploaded documents. The extracted text undergoes preprocessing using Natural Language Processing techniques such as tokenization, stop-word removal, and lemmatization. Relevant information including skills, education, work experience, certifications, and projects is identified and compared with the job description provided by the recruiter. A similarity algorithm calculates the degree of matching between the candidate profile and the job requirements. Finally, the system generates a resume score and ranks candidates according to their suitability, allowing recruiters to focus only on the most qualified applicants.

Chapter-7 : Technologies Used

The project is developed using Python because of its simplicity and extensive support for Artificial Intelligence and Machine Learning applications. Flask serves as the backend framework for developing the web application, while HTML, CSS, and JavaScript are used to design the user interface. MySQL stores candidate information, job descriptions, and matching results securely. Several Python libraries including spaCy, NLTK, Pandas, NumPy, and Scikit-learn are utilized for text preprocessing, skill extraction, data analysis, and similarity calculations. Libraries such as PyPDF2 and python-docx enable the system to extract text from PDF and Microsoft Word resumes.

Chapter-8 : Methodology

The system begins by accepting resumes uploaded by candidates in PDF or DOCX format. Once uploaded, the resume content is extracted and cleaned by removing unnecessary symbols, punctuation, and irrelevant words. Natural Language Processing techniques are then applied to tokenize the text, eliminate stop words, and convert words into their base forms using lemmatization. The processed information is analyzed to identify important entities such as technical skills, educational qualifications, certifications, and work experience. Simultaneously, the recruiter uploads a job description that undergoes the same preprocessing steps. Both the resume and job description are transformed into numerical vectors using the TF-IDF algorithm. Cosine Similarity is then applied to calculate the similarity score between the two vectors. Based on the resulting score, candidates are ranked in descending order of suitability, enabling recruiters to identify the best applicants efficiently.

Chapter-9 : Advantages

The Automated Resume Analyzer provides numerous advantages to both recruiters and job seekers. It significantly reduces the time required to evaluate resumes by automating repetitive tasks that traditionally required manual effort. The system improves the consistency and accuracy of candidate evaluation while minimizing unconscious bias during recruitment. It is capable of processing thousands of resumes within a short period, making it suitable for organizations conducting large-scale hiring. Furthermore, the ranking mechanism allows recruiters to identify the most qualified candidates quickly, leading to faster recruitment cycles and improved organizational productivity.

Chapter-10 : Future Scope

The project can be further enhanced by integrating advanced Artificial Intelligence technologies such as Large Language Models (LLMs) to improve semantic understanding of resumes and job descriptions. Future versions may include multilingual resume analysis, enabling the system to process resumes written in different languages. Additional features such as AI-powered interview scheduling, personalized resume improvement suggestions, voice and video resume analysis, blockchain-based certificate verification, and predictive hiring analytics can further enhance the system's capabilities. Cloud deployment and integration with major job portals would also improve accessibility and scalability.

Chapter-11 : Conclusion

The Automated Resume Analyzer for Job Portals represents an effective solution to the challenges faced during modern recruitment. By combining Artificial Intelligence, Natural Language Processing, and Machine Learning techniques, the system automates resume screening, improves candidate matching accuracy, and significantly reduces recruitment time. The proposed solution minimizes manual effort, enhances decision-making, and enables organizations to identify qualified candidates more efficiently. As recruitment continues to evolve with advances in AI, automated resume analysis systems will become an essential component of intelligent talent acquisition platforms.

Chapter-12 : References

The development of the Automated Resume Analyzer for Job Portals was based on information collected from various reliable sources, including research papers, books, and official documentation. Research papers published by IEEE, Springer, and Elsevier were studied to understand the concepts of Artificial Intelligence, Natural Language Processing, Machine Learning, resume parsing, and candidate ranking. The official Python Documentation was used to develop the application and implement the program efficiently. The spaCy Documentation helped in learning text processing techniques such as tokenization, lemmatization, and named entity recognition. The NLTK Documentation was referred to for text preprocessing methods like stop-word removal and stemming. The Scikit-learn Documentation was used to understand and implement machine learning techniques such as TF-IDF vectorization and Cosine Similarity for resume matching. These references provided the necessary theoretical knowledge and practical guidance for designing and implementing the proposed automated resume analyzer system.